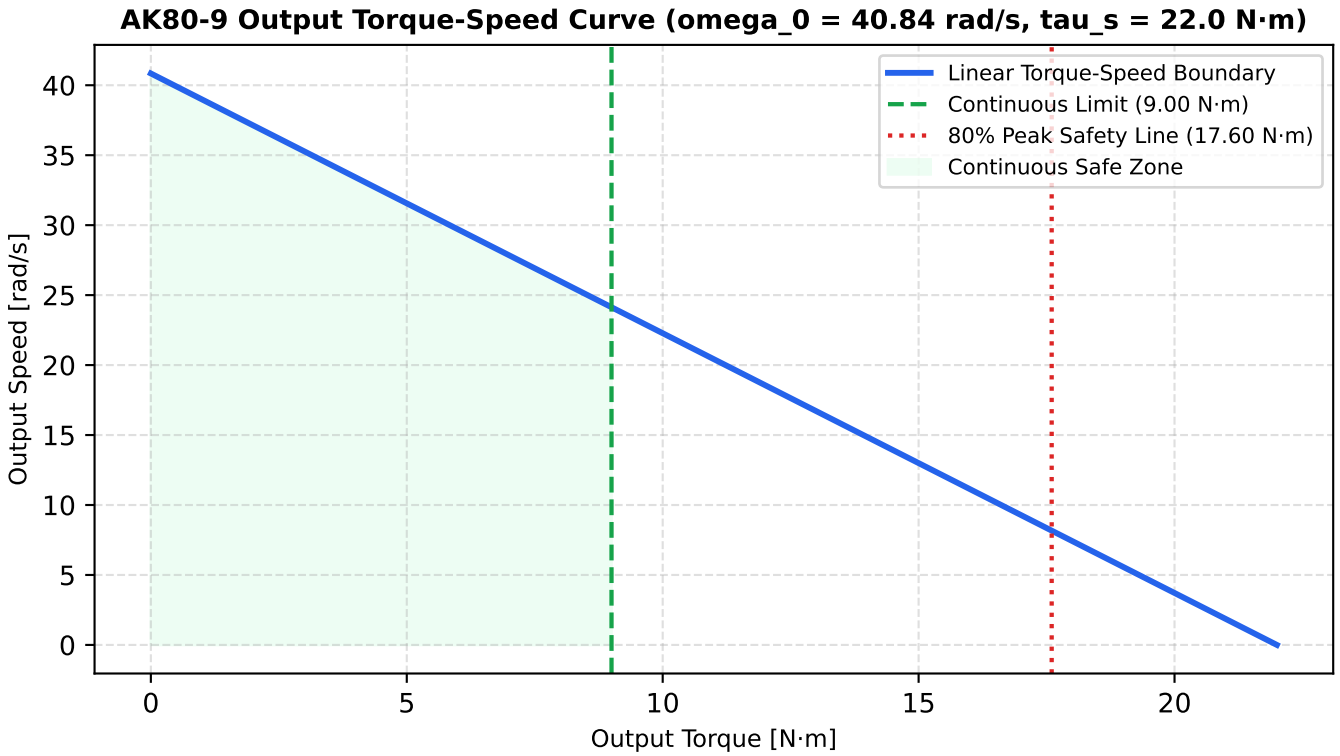
Part Number: AK80-9

1. Electrical & Mechanical Specifications (Standard Reference)

Parameter	Symbol	Specification Value	Test Condition / Notes
Part Number	ID	AK80-9	AK-Series QDD
Manufacturer	Vendor	CubeMars	Outer-Rotor BLDC QDD
Rated Input Voltage	V_rated	24.0 V	Nominal DC Bus
Stall / Peak Torque	tau_stall	22.0 N.m peak torque	At 24.0 V
Continuous Rated Torque	tau_cont	9.0 N.m rated torque	25 degC ambient, continuous duty
No-Load Speed	omega_0	390 rpm (at 24V KV100)	Unloaded output shaft
Actuator Mass (Weight)	m_act	490 g	Includes housing & internal gearhead
Internal Reduction	i_int	9 : 1 Planetary QDD	Integrated gear stage
Envelope Dimensions	W x H x D	88.0 x 40.5 x 88.0 mm	Outer housing envelope



AK80-9 — Mechanical Drawing & Operating Guidelines

2. Mechanical Mounting & Frame Integration
   - Housing Envelope: 88.0 mm x 40.5 mm x 88.0 mm
   - Total Verified Mass: 490 g (0.4900 kg)
   - Mounting Pattern: M2.5 / M3 ISO radial & horn flange threads

3. Thermal & Duty-Cycle Notes
   - Continuous operation must remain at or below the continuous rated torque.
   - If continuous torque is not explicitly tabulated on Page 1, apply a 0.33x stall torque
   - Peak torque operation (>80% of available speed-dependent torque) risks step loss or the

4. Citation Verification Hash
   - Part ID: AK80-9 | Source Document: ak80-9.pdf | Spec Table: Page 1